

# Cross-sectional factor backtester: methodology note

US large caps, 2010-2026. Generated 2026-09-15 from the saved runs; the repository README carries the same numbers.

## Question

Do value, momentum, quality and low volatility earn a premium on a universe an outsider can verify, once the fundamentals are lagged to their filing date, costs and a one-day execution lag are charged, and the number of specifications tried is counted honestly?

## Data, and what is honest about it

- Universe: S&P 500 membership reconstructed month by month from Wikipedia's changes table, cross-checked against an independent daily list (98.4% of universe-months agree, every disagreement listed) and 20 changes checked against S&P press releases. 496-504 names at every month-end, 818 unique names over the window.
- Fundamentals: SEC Financial Statement Data Sets, 2009q1-2026q2, 16 concepts through an ordered XBRL tag map. Every value is keyed on its filing date; the first-filed value wins over amendments; the value used at month-end t was filed on or before t - 1 day. This join has three unit tests, one of which plants a filing dated after the signal date and asserts it is excluded.
- Prices: yfinance daily adjusted closes. Delisted names are largely absent: 14.6% of universe-months have no price, 31% in 2010 falling to 0% today. yfinance's series for reused symbols are dropped by three logged rules (no history inside membership, bad print, corrupt series).
- Benchmarks: Ken French's factors and the six size x B/M and size x profitability portfolios, whose big-cap legs are the like-for-like comparison for a large-cap universe.

## Method

- Signals at each month-end from data available then: 12-1 momentum, book-to-price, earnings yield, gross profit / assets, cash-flow accruals (negated), 252-day volatility (negated), and the cosine similarity of the latest 10-K text to the prior year's (Cohen, Malloy and Nguyen 2020), from EDGAR primary documents dated by the SEC's filing date, scored without a language model so that nothing trained after the filing can leak into the score. Winsorised at the 1st/99th percentile, z-scored within sector (SIC mapped by hand to 11 GICS-like buckets), composites averaged and re-standardised.
- Portfolios: ten equal-count deciles, equal-weighted (cap-weighted as a sensitivity), formed monthly and traded at the close of the next trading day; long-short is decile 10 minus 1. Turnover is half the sum of absolute weight changes against drifted weights, both legs. Every dollar bought or sold pays the one-way cost c, so  $r_{net} = r_{gross} - 2c \times \text{turnover}$ ; the brief's  $r_{gross} - c \times \text{turnover}$  charges half that.
- Statistics: Spearman IC by month and by horizon 1-12 with an exponential half-life; Fama-MacBeth premia with Newey-West errors (checked against statsmodels to 1e-9); attribution on Mkt-RF, HML, UMD, RMW; break-even cost; and the deflated Sharpe with N read from a specification log that every run appends to (N = 329).

## Results (net of 10 bp; DSR uses N = 329)

| Factor                   | Gross | Net   | Vol   | Sharpe | DSR  | MaxDD | Turn. | IC     | IC t | Break-even |
|--------------------------|-------|-------|-------|--------|------|-------|-------|--------|------|------------|
| Momentum 12-1            | 2.1%  | 0.6%  | 15.5% | 0.04   | 0.00 | -56%  | 0.62  | 0.006  | 0.6  | 14 bp      |
| Value (B/P, E/P)         | -1.1% | -1.8% | 11.0% | -0.16  | 0.00 | -50%  | 0.30  | -0.004 | -0.5 | none       |
| Quality (GP/A, accruals) | 2.7%  | 2.1%  | 8.8%  | 0.24   | 0.02 | -24%  | 0.25  | 0.008  | 1.4  | 45 bp      |
| Low volatility           | -4.5% | -5.1% | 18.6% | -0.27  | 0.00 | -74%  | 0.25  | 0.003  | 0.2  | none       |
| Composite                | -3.6% | -4.7% | 14.1% | -0.33  | 0.00 | -66%  | 0.45  | -0.002 | -0.2 | none       |
| 10-K text similarity     | -0.2% | -1.0% | 5.6%  | -0.17  | 0.00 | -27%  | 0.30  | 0.004  | 0.9  | none       |

## Validation

Momentum long-short correlates 0.78 with UMD and loads on it with beta 0.87, R-squared 0.65: it is UMD, and UMD earned nothing in this window. The reported value and quality factors are sector-neutral composites and correlate 0.56 and 0.07 with HML and RMW, which is construction, not a join error. The join is tested by replicating French's construction (one signal, no sector neutralisation, cap-weighted terciles) against the big-cap half of his factor: the B/P replication correlates 0.78 with the big-cap HML leg over the window and 0.90 from 2016 (0.78 against full HML). The profitability replication reaches 0.62 against the big-cap leg and 0.62 against full RMW, a fail against the 0.6 bar; its early-years weakness was the ticker-to-CIK map (decisions/cik\_audit.md), not XBRL coverage.

## What the numbers mean

- Value nets -0.16 and loads 0.42 on HML (t 7.7) with an alpha of -1.5% (t -0.7): large-cap value earned nothing over the window. The 0.48 an earlier version of this note reported was a market-cap bug: split-adjusted prices met unadjusted share counts, so pre-split winners sat in the value long leg (decisions/f2\_shares.md).
- Momentum is marginal after costs (0.04; break-even 14 bp at 0.62 monthly turnover). Quality is the only positive line (0.24; IC t 1.4), carried by accruals. Low volatility loses as a long-short: beta -0.65 (t -10.7), RMW 0.92 (t 8.1), alpha 0.8% (t 0.3):

a short-beta position, and shorting beta lost for sixteen years. The composite nets -0.33.

- The deflated Sharpe of quality is 0.02 against all 329 logged rows and 0.06 against the 55 candidate rows: the probability that its Sharpe beats the best of that many random trials with the same dispersion. Nothing here is a 2.0 Sharpe.

### **What did not work**

- Wikipedia's 'Date added' is a corporate-event date for long-standing members (Sempra 2017, T. Rowe Price 2019); trusted only before 2010, adjudicated by the cross-check after.
- Stooq serves a JavaScript challenge, not data; yfinance stitches reused symbols and carries bad prints. Delisted names remain the survivorship that is left, and it is quantified.
- The first momentum run correlated 0.04 with UMD: uncleaned prices and a validation join off by one month. Both runs stay in the specification log.
- Wikipedia's CIK for ExxonMobil is a 2026 entity; OperatingIncomeLoss is not reported by banks. Each has a logged rule.
- The ticker-to-CIK map left 18% of members without filings (successor registrants, name collisions); 93 hand-verified override rows with a CIK per era fixed it. Market caps met a split-adjusted price with an unadjusted count; the value premium the first version reported was that bug. Both found by the September 2026 review, both in decisions/.

*Code, tests, the specification log and every check file are in the repository. The results table, charts and this note are regenerated by `backtester report`.*